# PROJECT REPORT

**Title:**

Online Payments Fraud Detection Using Machine Learning

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**Guide:** Santhosh Kumar

**College:** SVCE

**Academic Year**: 2025–2026

# 1. INTRODUCTION

## 1.1 Project Overview

Online payment fraud is a significant issue affecting financial institutions and customers worldwide. As digital transactions grow rapidly, fraudulent activities have become increasingly complex and challenging to detect using traditional methods. This project leverages machine learning algorithms to accurately detect fraudulent online payments, ensuring enhanced security and customer trust.

## 1.2 Purpose

The purpose of this project is to develop an intelligent fraud detection system that identifies potentially fraudulent transactions in real time using advanced ML models like Random Forest, Decision Tree, SVM, Extra Trees, and XGBoost.

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# 2. IDEATION PHASE

## 2.1 Problem Statement

Traditional fraud detection systems rely on static rules, making them ineffective against evolving fraud patterns. This leads to high false positives and negatives. The proposed system applies machine learning to improve detection accuracy and adaptability.

## 2.2 Empathy Map Canvas

**Users:** Bank customers and financial institutions.
**Needs:** Security, trust, and smooth transaction experience.
**Pains:** Unauthorized transactions, blocked legitimate payments.
**Gains:** Safer digital payments, reduced manual intervention.

## 2.3 Brainstorming

The team brainstormed approaches to improve fraud detection through automation, predictive analytics, and continuous learning models.

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# 3. REQUIREMENT ANALYSIS

## 3.1 Customer Journey Map

1. User initiates online payment.
2. Transaction details are analyzed by the ML model.
3. Model predicts if the transaction is fraudulent or genuine.
4. Fraudulent ones are flagged for review.

## 3.2 Solution Requirement

- Dataset from Kaggle (financial transactions)
- Python with ML libraries: scikit-learn, XGBoost
- Jupyter Notebook / VS Code
- System with at least 8GB RAM and modern CPU.

## 3.3 Data Flow Diagram

User → Transaction Input → Preprocessing → Model Prediction → Result Display

## 3.4 Technology Stack

Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Kaggle dataset.

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# 4. PROJECT DESIGN

## 4.1 Problem Solution Fit

Machine learning provides a scalable and adaptive solution for online payment fraud detection.

## 4.2 Proposed Solution

A predictive model that classifies transactions as “Fraud” or “Genuine.” It uses trained ML algorithms to detect anomalies in transaction patterns.

### 4.3 Solution Architecture

- **Input:** Transaction data
- **Process:** Preprocessing, model training, evaluation
- **Output:** Fraud prediction and accuracy metrics

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## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Milestone | Description |
|---|---|
| Milestone 1 | Project Initialization and Planning |
| Milestone 2 | Data Collection and Preprocessing |
| Milestone 3 | Model Development |
| Milestone 4 | Model Optimization and Tuning |
| Milestone 5 | Documentation and Demonstration |

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## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

Performance metrics such as Accuracy, Precision, Recall, and F1-score were calculated. The Random Forest model performed best, achieving high accuracy while minimizing false predictions.

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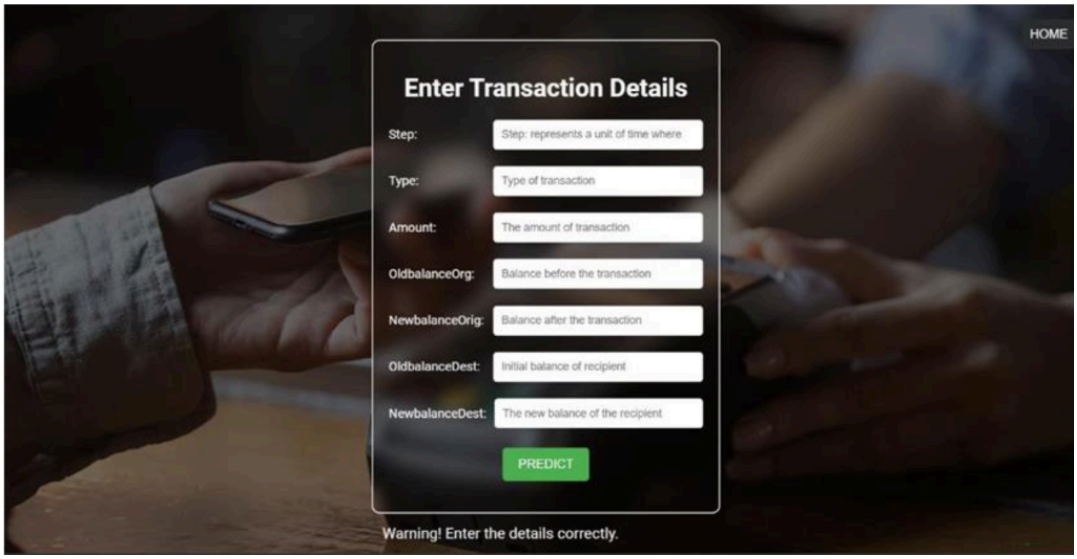
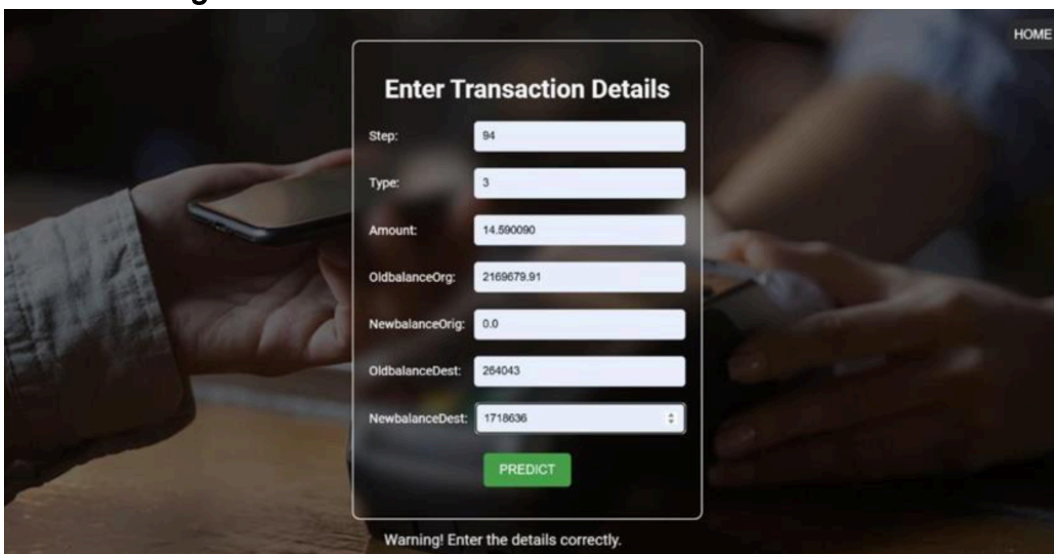
# 7. RESULTS

## 7.1 Output

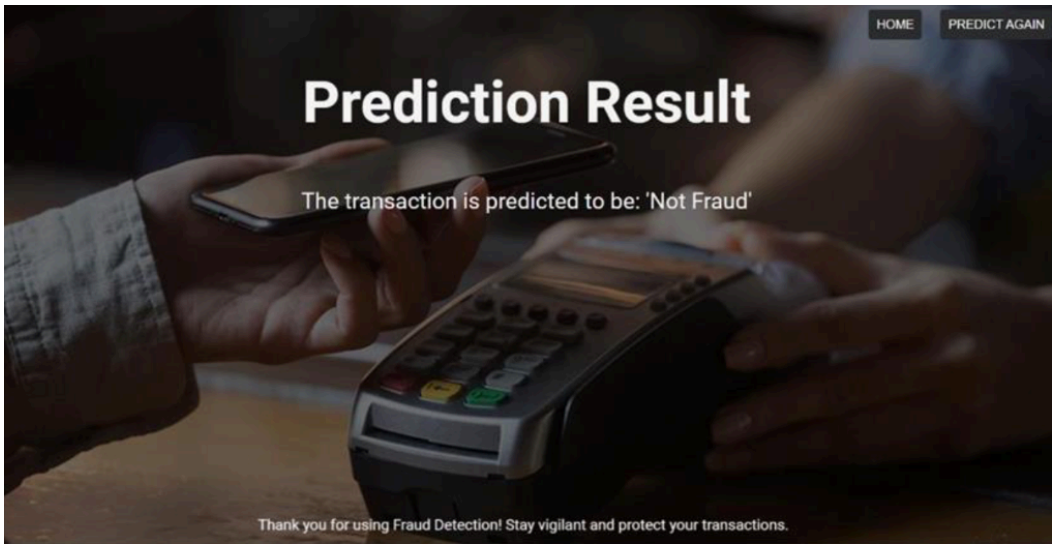
### Home Page:



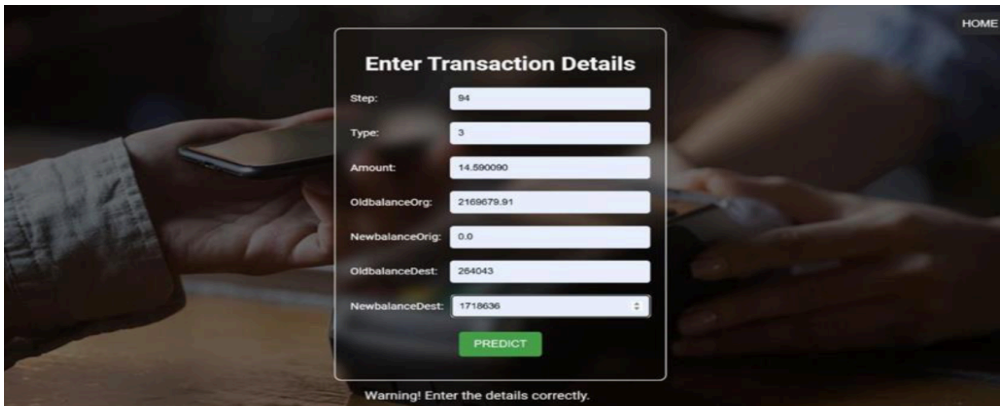

### Prediction Page:

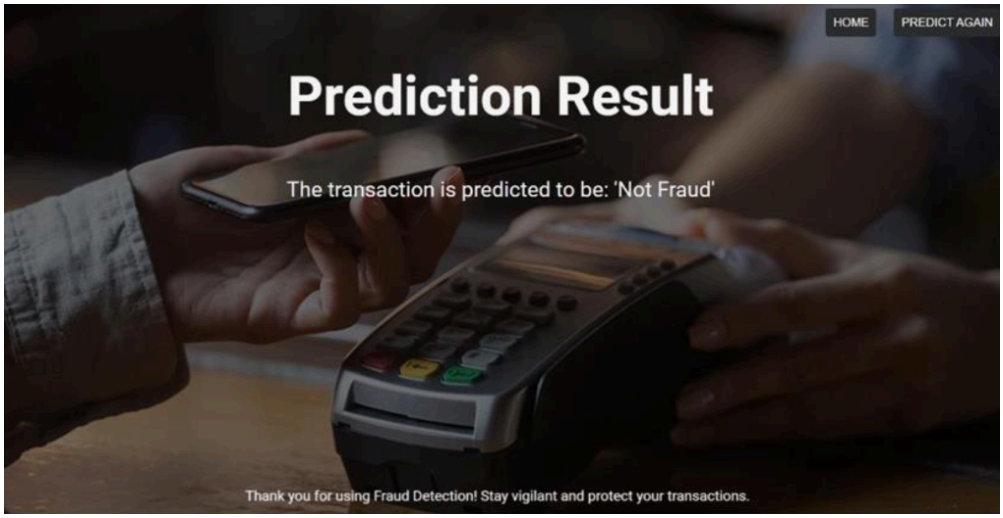



### Result Page:

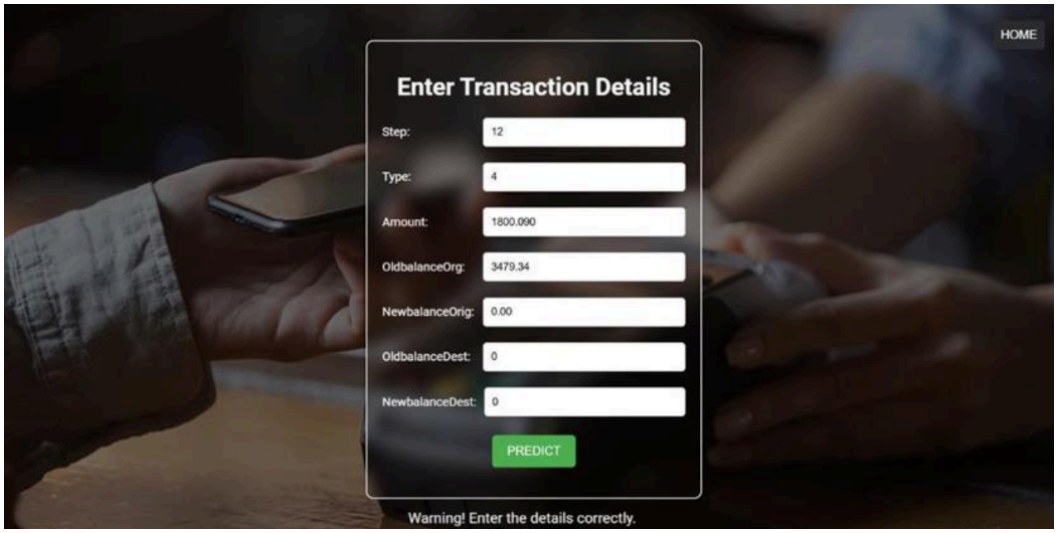


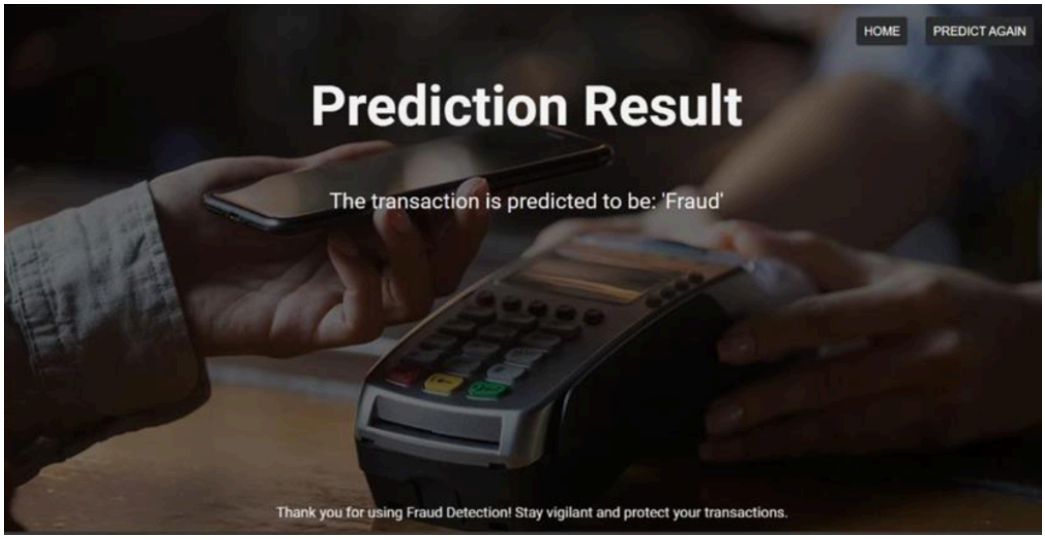

## Prediction 1:






## Prediction 2:



## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Real-time fraud detection
- Adaptive learning model
- Improved accuracy and reliability

### Disadvantages

- Requires large datasets for training
- Model retraining needed for new fraud trends

## 9. CONCLUSION

This project successfully demonstrates how machine learning can be used to identify fraudulent online payment transactions efficiently. The system improves over time with more data and provides financial institutions with a reliable method to prevent fraud.

## 10. FUTURE SCOPE

Future enhancements may include integrating deep learning models, real-time APIs, and visualization dashboards for fraud tracking.